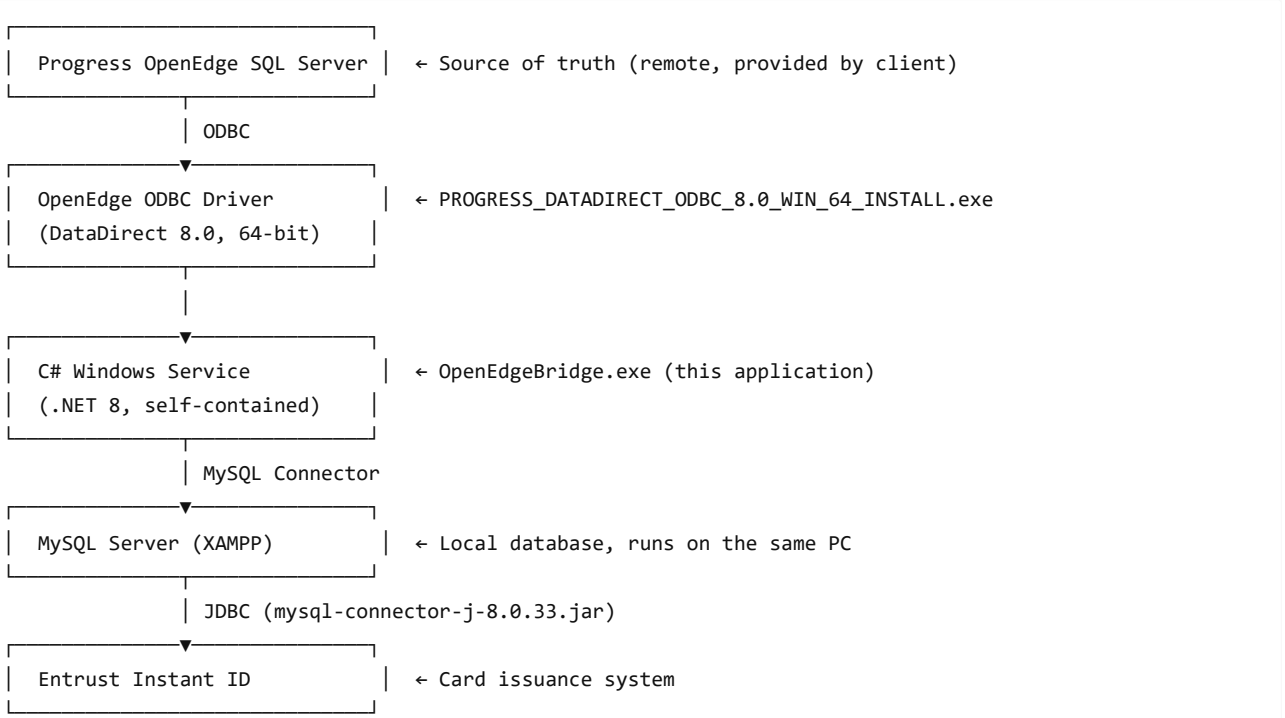
OpenEdge Bridge — Documentation

Version: 1.0
Date: July 23, 2026
Purpose: Sync bridge between Progress OpenEdge and MySQL for Entrust Instant ID card printing. Reads `"id-print"` from OpenEdge and mirrors it to MySQL's `id_print` table.

Architecture Overview



Data Flow

OpenEdge → MySQL (One-Way Sync)

```
OpenEdge "id-print" ──► C# Service ──► MySQL id_print table
       (read)              (sync loop)         (upsert)
```

All 24 fields from the `id-print` table are synced, including:

- `batch-no`, `batch-date`, `app-no`, `dob`
- `cp-no`, `fullname`, `hcp-fullname`, `address`, `hcp-cpno`
- `lname`, `fname`, `mname`
- `spa-docno`, `page-no`, `book-no`, `series-no`, `spa-atty`, `spa-dt-issue`
- `printed`, `printed-dt`, `printed-tm`
- `id-type`, `image-data`, `id-code`

Runs every N seconds (configurable in `appsettings.json`). Uses **INSERT ON DUPLICATE KEY UPDATE** to avoid duplicates. `print_count` is only inserted on first sync; subsequent updates do NOT overwrite it (Instant ID manages `print_count`).

MySQL → OpenEdge (Printed Back-Sync)

```
MySQL id_print ──────► C# Service ──────► OpenEdge "id-print"
 (print_count > 0)    (check + mark)       (SET printed = yes)
```

When Instant ID increments `print_count > 0` on a record, the bridge detects it and sets `printed = yes` in the corresponding OpenEdge record.

Prerequisites

On the Target Windows PC

#	Component	Version / File	Notes
1	**Windows OS**	Windows 10/11 or Windows Server 2016+	64-bit required
2	**OpenEdge ODBC Driver**	`PROGRESS_DATADIRECT_ODBC_8.0_WIN_64_INSTALL.exe`	Install first; configure a System DSN
3	**XAMPP**	Latest (8.x)	Provides local MySQL server
4	**MySQL Connector/J**	`mysql-connector-j-8.0.33.jar`	For Entrust Instant ID to connect to MySQL
5	**OpenEdgeBridge**	Published `publish/` folder	Self-contained, no .NET runtime needed

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Local Development — Installing Progress OpenEdge Database

> ***Use this section when no remote OpenEdge server is available.***
> *You can install Progress OpenEdge locally, create a database, and run the SQL below.*

Step 0-A: Download & Install Progress OpenEdge

1. Go to https://www.progress.com/openedge/downloads
2. Download **Progress OpenEdge Developer Studio** or **Progress OpenEdge Personal Edition** (free for development)
3. Run the installer (typically `PROGRESS_OE_12.x_WIN_64.exe`) as Administrator
4. Choose **Typical** or **Complete** installation
5. Default install path: `C:\Progress\OpenEdge\`

After installation, a **"Proenv"** shortcut will appear in your Start Menu.
Proenv is a command prompt pre-configured with all OpenEdge environment variables.

Step 0-B: Create a Local OpenEdge Database

Open **Proenv** as Administrator and run:

```
:: Create a folder for your database
mkdir C:\OpenEdgeDatabases
cd C:\OpenEdgeDatabases

:: Create an empty database called 'iddb'
prodb iddb empty
```

This creates `iddb.db` and related files in `C:\OpenEdgeDatabases\`.

Step 0-C: Start the Database Server

In **Proenv**, run:

```
:: Start the database for multi-user access with both 4GL and SQL enabled
proserve "C:\OpenEdgeDatabases\iddb.db" -S 11918 -ServerType Both -n 51 -L 8192
```

Parameter	Meaning
`-S 11918`	Service port (SQL and 4GL clients connect on this port)
`-n 51`	Maximum concurrent users
`-L 8192`	Lock table size (entries)
`-ServerType Both`	**Required** — enables both 4GL and SQL-92 engine

The database is now running. Keep this Proenv window open.

Step 0-D: Configure the ODBC System DSN

1. Press `Win + R`, type `odbcad32.exe`, press Enter
2. Go to the **System DSN** tab, click **Add...**
3. Select **Progress OpenEdge 12.x Driver** (or "DataDirect OpenEdge SQL Server")
4. Configure:
   - **Data Source Name:** `OpenEdgeInstantID`
   - **Description:** `OpenEdge Instant ID Database`
   - **Host Name:** `localhost`
   - **Port Number:** `11918`
   - **Database Name:** `iddb`
   - **User ID:** `admin`
5. Click **Test Connect** → should succeed
6. Click **OK**

Step 0-E: Create the Database Schema (`.df` File)

The table schema is defined in a `.df` (Data Definition) file. Create a file named `iddb.df` in `C:\OpenEdgeDatabases\` with the following content:

```
ADD TABLE "id-print"
  AREA "Schema Area"
  DUMP-NAME "id-print"

ADD FIELD "batch-no" OF "id-print" AS integer
  FORMAT "->,>>>,>>9"
  INITIAL "0"
  POSITION 2
  MAX-WIDTH 4
  ORDER 10

ADD FIELD "batch-date" OF "id-print" AS date
  FORMAT "99/99/9999"
  INITIAL ?
  POSITION 3
  MAX-WIDTH 4
  ORDER 20

ADD FIELD "app-no" OF "id-print" AS integer
  FORMAT "->,>>>,>>9"
  INITIAL "0"
```

```
  POSITION 4
  MAX-WIDTH 4
  ORDER 30

ADD FIELD "dob" OF "id-print" AS date
  FORMAT "99/99/9999"
  INITIAL ?
  POSITION 5
  MAX-WIDTH 4
  ORDER 40

ADD FIELD "cp-no" OF "id-print" AS character
  FORMAT "x(15)"
  INITIAL ""
  POSITION 6
  MAX-WIDTH 30
  ORDER 50

ADD FIELD "fullname" OF "id-print" AS character
  FORMAT "x(65)"
  INITIAL ""
  POSITION 7
  MAX-WIDTH 130
  ORDER 60

ADD FIELD "hcp-fullname" OF "id-print" AS character
  FORMAT "x(65)"
  INITIAL ""
  POSITION 8
  MAX-WIDTH 130
  ORDER 70

ADD FIELD "address" OF "id-print" AS character
  FORMAT "x(65)"
  INITIAL ""
  POSITION 9
  MAX-WIDTH 130
  ORDER 80

ADD FIELD "hcp-cpno" OF "id-print" AS character
  FORMAT "x(15)"
  INITIAL ""
  POSITION 10
  MAX-WIDTH 30
  ORDER 90

ADD FIELD "lname" OF "id-print" AS character
  FORMAT "x(55)"
  INITIAL ""
  POSITION 11
  MAX-WIDTH 110
  ORDER 100

ADD FIELD "fname" OF "id-print" AS character
  FORMAT "x(55)"
  INITIAL ""
```

```
  POSITION 12
  MAX-WIDTH 110
  ORDER 110

ADD FIELD "mname" OF "id-print" AS character
  FORMAT "x(55)"
  INITIAL ""
  POSITION 13
  MAX-WIDTH 110
  ORDER 120

ADD FIELD "spa-docno" OF "id-print" AS character
  FORMAT "x(25)"
  INITIAL ""
  POSITION 14
  MAX-WIDTH 50
  ORDER 130

ADD FIELD "page-no" OF "id-print" AS character
  FORMAT "x(20)"
  INITIAL ""
  POSITION 15
  MAX-WIDTH 40
  ORDER 140

ADD FIELD "book-no" OF "id-print" AS character
  FORMAT "x(20)"
  INITIAL ""
  POSITION 16
  MAX-WIDTH 40
  ORDER 150

ADD FIELD "series-no" OF "id-print" AS character
  FORMAT "x(20)"
  INITIAL ""
  POSITION 17
  MAX-WIDTH 40
  ORDER 160

ADD FIELD "spa-atty" OF "id-print" AS character
  FORMAT "x(65)"
  INITIAL ""
  POSITION 18
  MAX-WIDTH 130
  ORDER 170

ADD FIELD "spa-dt-issue" OF "id-print" AS date
  FORMAT "99/99/9999"
  INITIAL ?
  POSITION 19
  MAX-WIDTH 4
  ORDER 180

ADD FIELD "printed" OF "id-print" AS logical
  FORMAT "yes/no"
  INITIAL "no"
```

```
  POSITION 20
  MAX-WIDTH 1
  ORDER 190

ADD FIELD "printed-dt" OF "id-print" AS date
  FORMAT "99/99/9999"
  INITIAL ?
  POSITION 21
  MAX-WIDTH 4
  ORDER 200

ADD FIELD "printed-tm" OF "id-print" AS character
  FORMAT "x(35)"
  INITIAL ""
  POSITION 22
  MAX-WIDTH 70
  ORDER 210

ADD FIELD "id-type" OF "id-print" AS integer
  DESCRIPTION "1 = for RTC, 2=senior"
  FORMAT "->,>>>,>>9"
  INITIAL "0"
  POSITION 23
  MAX-WIDTH 4
  ORDER 220

ADD FIELD "image-data" OF "id-print" AS raw
  FORMAT "x(55)"
  INITIAL ""
  POSITION 24
  MAX-WIDTH 110
  ORDER 230

ADD FIELD "id-code" OF "id-print" AS integer
  FORMAT "->,>>>,>>9"
  INITIAL "0"
  POSITION 25
  MAX-WIDTH 4
  ORDER 240


.
PSC
cpstream=ISO8859-1
.
0000003166
```

Then, in **Proenv**, load the `.df` file into the database:

```
cd C:\OpenEdgeDatabases
prodict iddb
```

In the Data Dictionary menu, select **Admin → Load Data Definitions → .df file**, then choose `iddb.df`.

> ***Note:*** *The database name is* `iddb` *and the schema resides under the* `PUB` *area (default).*

Database Tables

`id-print` / `id_print`

> *Converted from the provided `.df` schema.*
> ***Paste each CREATE TABLE directly into your SQL editor and run it.***

OpenEdge SQL — CREATE TABLE

```
CREATE TABLE "id-print" (
    "batch-no"      INTEGER       NOT NULL DEFAULT 0,
    "batch-date"    DATE          NULL,
    "app-no"        INTEGER       NOT NULL DEFAULT 0,
    "dob"           DATE          NULL,
    "cp-no"         VARCHAR(15)   NOT NULL DEFAULT '',
    "fullname"      VARCHAR(65)   NOT NULL DEFAULT '',
    "hcp-fullname"  VARCHAR(65)   NOT NULL DEFAULT '',
    "address"       VARCHAR(65)   NOT NULL DEFAULT '',
    "hcp-cpno"      VARCHAR(15)   NOT NULL DEFAULT '',
    "lname"         VARCHAR(55)   NOT NULL DEFAULT '',
    "fname"         VARCHAR(55)   NOT NULL DEFAULT '',
    "mname"         VARCHAR(55)   NOT NULL DEFAULT '',
    "spa-docno"     VARCHAR(25)   NOT NULL DEFAULT '',
    "page-no"       VARCHAR(20)   NOT NULL DEFAULT '',
    "book-no"       VARCHAR(20)   NOT NULL DEFAULT '',
    "series-no"     VARCHAR(20)   NOT NULL DEFAULT '',
    "spa-atty"      VARCHAR(65)   NOT NULL DEFAULT '',
    "spa-dt-issue"  DATE          NULL,
    "printed"       BIT           NOT NULL DEFAULT 0,
    "printed-dt"    DATE          NULL,
    "printed-tm"    VARCHAR(35)   NOT NULL DEFAULT '',
    "id-type"       INTEGER       NOT NULL DEFAULT 0,
    "image-data"    VARBINARY(110) NULL,
    "id-code"       INTEGER       NOT NULL DEFAULT 0
);
```

> ***Note:*** *OpenEdge requires double-quotes around field names containing hyphens (`-`).*
> *The schema resides in the `PUB` area (default).*
> *`BIT` represents the OpenEdge `LOGICAL` type (yes/no → 1/0).*
> *`VARBINARY(110)` stores the raw image binary data (MAX-WIDTH 110 from the schema).*

MySQL — CREATE TABLE

```
CREATE TABLE IF NOT EXISTS id_print (
    id              INT           NOT NULL AUTO_INCREMENT,
    batch_no        INT           NULL,
    batch_date      DATE          NULL,
    app_no          INT           NULL,
    dob             DATE          NULL,
    cp_no           VARCHAR(15)   NULL,
    fullname        VARCHAR(65)   NULL,
    hcp_fullname    VARCHAR(65)   NULL,
    address         VARCHAR(65)   NULL,
    hcp_cpno        VARCHAR(15)   NULL,
    lname           VARCHAR(55)   NULL,
```

```
    fname          VARCHAR(55)   NULL,
    mname          VARCHAR(55)   NULL,
    spa_docno      VARCHAR(25)   NULL,
    page_no        VARCHAR(20)   NULL,
    book_no        VARCHAR(20)   NULL,
    series_no      VARCHAR(20)   NULL,
    spa_atty       VARCHAR(65)   NULL,
    spa_dt_issue   DATE          NULL,
    printed        VARCHAR(3)    NULL        COMMENT 'yes / no (matches OpenEdge LOGICAL FORMAT)',
    printed_dt     DATE          NULL,
    printed_tm     VARCHAR(35)   NULL,
    id_type        INT           NULL        COMMENT '1 = for RTC, 2 = senior',
    image_data     LONGBLOB      NULL,
    id_code        INT           NULL,
    print_count    INT           NULL        COMMENT 'Incremented by Instant ID; triggers printed=yes
back-sync',
    PRIMARY KEY (id),
    UNIQUE INDEX idx_app_no (app_no)
) ENGINE=InnoDB
  DEFAULT CHARSET=utf8mb4
  COLLATE=utf8mb4_unicode_ci;
```

> ***Note:*** *`id` is an internal auto-increment primary key. `app_no` is the unique business key — the bridge uses it for deduplication (`ON DUPLICATE KEY UPDATE` fires when the same `app_no` is synced again).*

Step-by-Step Setup (Remote Server)

> ***If you already have a remote OpenEdge server**, follow these steps.*
> ***If no server is available**, go back to [Local Development — Installing Progress OpenEdge Database] first.*

1. Install OpenEdge ODBC Driver

Run the installer as Administrator:

```
PROGRESS_DATADIRECT_ODBC_8.0_WIN_64_INSTALL.exe
```

After installation, open **ODBC Data Sources (64-bit)** and create a **System DSN**:

1. Press `Win + R`, type `odbcad32.exe`, press Enter
2. Go to the **System DSN** tab
3. Click **Add...**
4. Select **Progress DataDirect OpenEdge SQL Server** (or similar)
5. Configure:
   - **Data Source Name:** e.g., `OpenEdgeInstantID`
   - **Host:** (provided by your OpenEdge team, or `localhost` if local)
   - **Port:** (provided by your OpenEdge team, or `11918` if local)
   - **Database:** (provided by your OpenEdge team, or `iddb` if local)
   - **User ID / Password:** (provided by your OpenEdge team)
6. Click **Test Connect** to verify, then **OK**

2. Create the OpenEdge Tables

Connect to your OpenEdge database using any ODBC-compatible SQL client (DBeaver, HeidiSQL, etc.) and paste the SQL from the Database Tables section above:

- Run the `"id-print"` CREATE TABLE

3. Install XAMPP

1. Download from https://www.apachefriends.org/
2. Run the installer with default settings
3. Open the **XAMPP Control Panel**
4. Click **Start** next to **MySQL**
5. Click **Admin** to open phpMyAdmin

4. Create the MySQL Database & Tables

In phpMyAdmin (or any MySQL client), run:

```
CREATE DATABASE IF NOT EXISTS instantid_bridge
  CHARACTER SET utf8mb4
  COLLATE utf8mb4_unicode_ci;

USE instantid_bridge;
```

Then paste and run the MySQL CREATE TABLE statements from the Database Tables section above:

- Run the `id_print` CREATE TABLE

5. Configure Entrust Instant ID (MySQL Connector/J)

1. Copy `mysql-connector-j-8.0.33.jar` to Entrust's `lib/` folder
2. Configure Entrust to point to the local MySQL database:
   - **Host:** `localhost`
   - **Port:** `3306`
   - **Database:** `instantid_bridge`
   - **User:** `root` (or a dedicated user)
   - **Table:** `id_print`

6. Deploy the Bridge Service

Copy the entire `publish/` folder to the target PC, e.g. `C:\OpenEdgeBridge\` .

Edit `appsettings.json`

```
{
  "ConnectionStrings": {
    "OpenEdge": "DSN=OpenEdgeInstantID;Uid=your_user;Pwd=your_password;",
    "MySql": "Server=localhost;Database=instantid_bridge;Uid=root;Pwd=your_mysql_password;"
  },
  "Sync": {
    "IntervalSeconds": 10
  }
}
```

Setting	Description
`OpenEdge`	ODBC DSN name, user, and password
`MySql`	MySQL server, database, user, and password
`Sync:IntervalSeconds`	How often the sync loop runs (default: 10s)

7. Install as a Windows Service

Open **PowerShell as Administrator** and run:

```
New-Service -Name "OpenEdgeBridge" `
            -BinaryPathName "C:\OpenEdgeBridge\OpenEdgeBridge.exe" `
            -DisplayName "OpenEdge Bridge Service" `
            -Description "Syncs id-print data between OpenEdge and MySQL for Entrust Instant ID" `
            -StartupType Automatic

Start-Service OpenEdgeBridge
```

Verify:

```
Get-Service OpenEdgeBridge
# Status should be: Running
```

Service Management Commands

```
Start-Service OpenEdgeBridge      # Start
Stop-Service OpenEdgeBridge       # Stop
Restart-Service OpenEdgeBridge    # Restart (after config changes)
Get-Service OpenEdgeBridge        # Check status

# Uninstall
Stop-Service OpenEdgeBridge
sc delete OpenEdgeBridge
```

Logging & Troubleshooting

Log Locations

Source	Location
File logs	`C:\OpenEdgeBridge\Logs\` (dated log files)
Windows Event Viewer	*Windows Logs → Application*, Source: `OpenEdgeBridge`

Common Issues

Problem	Likely Cause	Fix
Service starts then stops immediately	Wrong DSN name or credentials	Test the ODBC DSN in ODBC Administrator with **Test Connect**
"MySQL connection string not found"	`appsettings.json` not in same folder as `.exe`	Ensure `appsettings.json` is in `C:\OpenEdgeBridge\`
Duplicate rows in MySQL	Missing UNIQUE index	Run the `ALTER TABLE ... ADD UNIQUE INDEX` SQL above
OpenEdge ODBC driver not found	Wrong bitness (32 vs 64)	Use `odbcad32.exe` (64-bit), not the 32-bit one in SysWOW64
`Failure getting schema read lock (12138)`	SQL engine not initialized for database	1. Verify `-ServerType Both` or `-ServerType SQL` on proserve 2. In sqlexp run: `UPDATE TABLE STATISTICS AND INDEX`

		`STATISTICS AND ALL COLUMN STATISTICS;` 3. If still failing, restart DB: `proshut` then `proserve` with `-SQLTrc 3`
MySQL not accessible from Entrust	MySQL not running or wrong port	Check XAMPP Control Panel; MySQL must be **Started**
OpenEdge SQL CREATE TABLE fails with "Invalid table name"	Hyphens in table/field names not quoted	Wrap names containing hyphens in double quotes: `"id-print"`, `"batch-no"`
`image-data` field too small	VARBINARY length insufficient	The OpenEdge schema specifies MAX-WIDTH 110; adjust as needed

---

Sync Interval Tuning

The default sync interval is **10 seconds**. Adjust in `appsettings.json`:

```
"Sync": {
  "IntervalSeconds": 30
}
```

- **Lower** (e.g., 5s) = faster sync, more database load
- **Higher** (e.g., 60s) = slower sync, less load
- Recommended: 10–30 seconds for production

Port Summary

Component	Port	Direction
OpenEdge SQL Server	Provided by OpenEdge team	Outbound
MySQL (XAMPP)	3306	Local
Entrust → MySQL	3306	Local via JDBC

No inbound ports are opened by the bridge itself.

Upgrade Procedure

1. Stop the service: `Stop-Service OpenEdgeBridge`
2. Replace all files in `C:\OpenEdgeBridge\` with the new `publish/` output
3. Preserve your `appsettings.json` (do not overwrite with default)
4. Start the service: `Start-Service OpenEdgeBridge`

Checklist for Target PC

- [ ] Windows 10/11 or Server 2016+ (64-bit)
- [ ] OpenEdge ODBC driver installed and DSN configured
- [ ] XAMPP installed and MySQL running
- [ ] `instantid_bridge` database created in MySQL
- [ ] `id-print` / `id_print` table created in **both** OpenEdge and MySQL
- [ ] `mysql-connector-j-8.0.33.jar` placed in Entrust's lib folder
- [ ] Bridge `publish/` folder copied to `C:\OpenEdgeBridge\`

- [ ] `appsettings.json` updated with correct connection strings
- [ ] Windows Service installed and started
- [ ] Logs checked for errors